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*Engineering a Centralized Web Portal for Data
Center Virtual Machine Migration Tracking*

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Abstract / Résumé

Abstract

In large-scale enterprise infrastructures, deploying Virtual Machines (VMs) and publishing web services demand coordination between technical application teams and cybersecurity governance. Within SONATRACH's Pipeline Transport Activity (*Activité Transport par Canalisation* - TRC), processing infrastructure publication requests historically relied on manual multi-tab spreadsheet exchanges ([formulaire_création_VM.xlsx](#)). This manual workflow introduced data formatting errors, missing network protocol dependencies, and prolonged security validation cycles. This project introduces an automated web portal featuring an embedded intelligent recommendation engine. Developed under a 3-tier architecture, the platform guides users through a wizard-based interface, leverages a rule-based inference engine to automatically populate network firewall matrices based on selected software stacks, enforces Role-Based Access Control (RBAC) for security audits, and dynamically generates multi-tab Excel spreadsheets. System evaluation demonstrates 100% rule-matching accuracy for standard software stacks and a 96% parsing accuracy benchmark on legacy datasets.

Keywords: Virtual Machine Provisioning, Intelligent Assistant, Network Flow Matrix, Rule Engine, Process Automation. (*Drop: SONATRACH TRC*).

Résumé

Dans les infrastructures d'entreprise à grande échelle, le déploiement de machines virtuelles (VM) et la publication de services web nécessitent une coordination rigoureuse entre les équipes applicatives et la gouvernance de la cybersécurité. Au sein de l'Activité Transport par Canalisation (TRC) de SONATRACH, le traitement des demandes de publication d'infrastructure reposait historiquement sur des échanges manuels de fichiers tableurs multi-onglets ([formulaire_création_VM.xlsx](#)). Ce processus manuel engendrait des erreurs de formatage, des omissions dans la matrice des flux réseau et des retards dans la validation sécuritaire. Ce projet présente un portail web automatisé doté d'un moteur de recommandation intelligent. Développée selon une architecture à trois tiers, la plateforme guide l'utilisateur via une interface pas-à-pas, exploite un moteur d'inférence à règles pour générer automatiquement les flux réseau selon les composants logiciels sélectionnés, applique un contrôle d'accès basé sur les rôles (RBAC) pour l'audit de sécurité, et génère dynamiquement des fichiers Excel multi-onglets conforme. L'évaluation du système montre une précision de 100% dans la suggestion des flux et un taux de réussite de 96% lors de l'ingestion de formulaires historiques.

Mots-clés : Publication de Machines Virtuelles, Assistant Intelligent, Matrice des Flux Réseau, Moteur de Règles, Automatisation de Processus. (*Drop: SONATRACH TRC*).

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List of Abbreviations

- 2CS: 2ème Année Cycle Supérieur (Second Year Master's Level);
- 3NF: Third Normal Form;
- VM: Virtual Machine;
- API: Application Programming Interface;
- CRUD: Create, Read, Update, Delete;
- DMZ: Demilitarized Zone;
- DNS: Domain Name System;
- DTI: Direction des Technologies et de l'Information ;
- ERD: Entity-Relationship Diagram;
- ERP: Enterprise Resource Planning;
- HCI: Human-Computer Interaction;
- HTTP: Hypertext Transfer Protocol;
- HTTPS: Hypertext Transfer Protocol Secure;
- IP / IPv4 : Internet Protocol (Version 4) ;
- JSON: JavaScript Object Notation;
- JWT: JSON Web Token;
- LDAP: Lightweight Directory Access Protocol;
- LLM: Large Language Model;
- LPG: Liquefied Petroleum Gas;
- NGL: Natural Gas Liquids;
- ORM: Object-Relational Mapping;
- OS: Operating System;
- PERN: PostgreSQL, Express.js, React.js, Node.js Stack;
- POLP: Principle of Least Privilege;
- PSSI: Politique de Sécurité des Systèmes d'Information ;
- RBAC: Role-Based Access Control;
- RDP: Remote Desktop Protocol;
- REST: Representational State Transfer;
- SD: Systèmes Intelligents et Données (Intelligent Systems & Data) ;
- SPE: Stage Pré-Embauche (Pre-Employment Internship);
- SSH: Secure Shell;
- SSI / SI: Sécurité des Systèmes d'Information / Système d'Information ;
- SSO: Single Sign-On;
- TLS: Transport Layer Security;
- TRC: Transport par Canalisation (Sonatrach TRC Activity) ;
- UI: User Interface;
- UML: Unified Modeling Language.

Introduction

In enterprise IT infrastructure management, provisioning and publishing Virtual Machines (VMs) are fundamental operational tasks requiring coordination between application teams and cybersecurity teams. Publishing an enterprise application involves establishing network firewall rules, routing traffic through Demilitarized Zones (DMZs), assigning IPv4 addresses, configuring server runtime dependencies, and conducting security audits.

Within SONATRACH's Pipeline Transport Activity (*Activité Transport par Canalisation - TRC*), managing VM publication requests relied on manual Microsoft Excel workbooks ([formulaire_création_VM.xlsx](#)). Requesters manually supplied contact information, hardware parameters, software components, network flow matrices, and compliance tracking items across four distinct worksheets. Because non-specialist applicants frequently struggled to define network dependencies (such as omitting port 1521 for Oracle databases or port 5432 for PostgreSQL), submissions suffered from input syntax errors and missing firewall rules. Consequently, requests underwent multiple email revision loops between applicants and the IT Security (*Sécurité SI*) team, leading to operational delays.

To address these challenges, this project delivers an automated, web-based VM publication portal integrated with an intelligent recommendation engine. The system digitizes request creation through a guided step-by-step submission wizard, leverages an inference engine to automatically infer and generate required network flow rules based on selected software stacks, enforces strict Role-Based Access Control (RBAC) for security sign-offs, and dynamically compiles validated requests into compliant multi-tab Excel spreadsheets.

Chapter 1: Project Context and Host Organization

1.1. Presentation of SONATRACH Group

SONATRACH (*Société Nationale pour la Recherche, la Production, le Transport, la Transformation, et la Commercialisation des Hydrocarbures*) is Algeria's national state-owned energy enterprise and the largest hydrocarbon corporation in Africa. Managing the complete energy value chain, SONATRACH operates across upstream exploration, crude oil and natural gas extraction, downstream refining, petrochemical production, international shipping, and pipeline transportation networks.

1.2. The Pipeline Transport Activity (TRC)

The *Activité Transport par Canalisation* (TRC) is the primary operating division of SONATRACH responsible for managing, operating, and maintaining Algeria's hydrocarbon transport pipeline network. TRC oversees thousands of kilometers of pipelines carrying crude oil, natural gas, liquefied petroleum gas (LPG), and condensate from extraction fields in the South to coastal export terminals and domestic refineries. Its operational footprint spans major regional hubs including Algiers, Oran, Skikda, and Hassi Messaoud. Maintaining operational continuous monitoring and telemetric data processing across these sites requires resilient, high-availability server infrastructure.

1.3. Presentation of the Host Department

This project was carried out within the IT and Information Technologies Department (*Direction des Technologies et de l'Information - DTI*) of the TRC activity. The DTI is structured into three specialized operational units:

- **Virtualization & Infrastructure Unit:** Provisions hypervisors, manages data center compute clusters, and allocates virtual machine resources.
- **Network & Cybersecurity Unit (Sécurité SI):** Defines network segmentation, manages perimeter firewall rules, configures DMZs, conducts vulnerability scans, and enforces security compliance.
- **Software Engineering & Digitalization Unit:** Designs, deploys, and maintains enterprise web applications, automating internal business processes and administrative workflows.

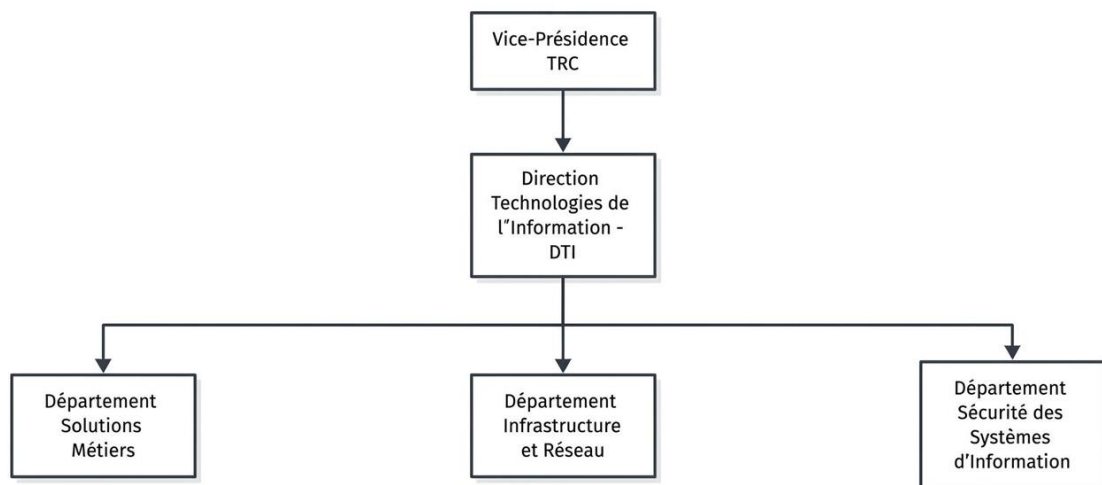


Figure 1: DTI Organizational Hierarchy

1.4. Project Context and Problem Statement

Publishing a virtual machine or enterprise web application within TRC requires gathering technical specifications across four main domains:

1. **Applicant Metadata:** Organizational division (Pôle), structure, service manager, and technical contact details.
2. **VM Technical Parameters:** Service name, target user population, DNS hostnames, primary IP address allocations, operating system, and software stack components (e.g., Apache, Nginx, Node.js, Python, Oracle, PostgreSQL).
3. **Network Flow Matrix:** Detailed source IP, destination IP, communication protocol, port numbers, flow classification (web, database, administration), and functional description.
4. **Security Non-Conformity Tracking:** DMZ isolation verification, antivirus deployment, TLS certificate configurations, vulnerability scan results, and code review approvals.

1.5. Problem Statement

Relying on standalone Excel spreadsheets for this workflow creates three operational bottlenecks:

- **Syntax and Data Entry Errors:** Requesters manually enter IP subnets, ports, and hostnames without real-time validation, causing invalid submissions.
- **Incomplete Network Flow Matrices:** Requesters often lack specialized networking knowledge and omit required firewall rules (e.g., missing SSH port 22 for Linux, RDP port 3389 for Windows, or port 1521 for Oracle connections), causing rejections by the *Sécurité SI* team.
- **Fragmented Security Auditing:** Updating security compliance fields within unencrypted Excel files via email provides no centralized audit trail or access control, creating governance risks.

The objective of this project is to build an automated, web-based VM publication application featuring an intelligent recommendation engine that infers required network flows, validates input syntax in real time, enforces role-based access for security reviews, and generates fully compliant multi-tab Excel files ([formulaire_création_VM.xlsx](#)).

Chapter 2: Analysis of the Existing System and Requirements Elaboration

2.1. Analysis of the Existing System and Work Processes

The legacy VM publication process at TRC relied entirely on emailing a 4-tab Microsoft Excel workbook ([formulaire_création_VM.xlsx](#)). The process followed four steps:

[Applicant] Fills Tabs 1, 2 & 3 in Excel ---> [Email Transmission] ---> [Sécurité SI] Audits & Fills Tab 4 ---> [Final Approval / Rejection]

1. **Form Completion:** The applicant manually filled Tab 1 (*Principale*), Tab 2 (*Publication VM*), and Tab 3 (*Informations liées au service*).
2. **File Handoff:** The completed file was sent via email to the *Sécurité SI* team.
3. **Security Audit:** Security officers evaluated network matrix rows, performed manual port verification, conducted vulnerability scans, and updated Tab 4 (*Suivi des non-conformités*).
4. **Iterative Rejection Loop:** If errors or missing rules were detected, the file was sent back to the requester for manual correction.

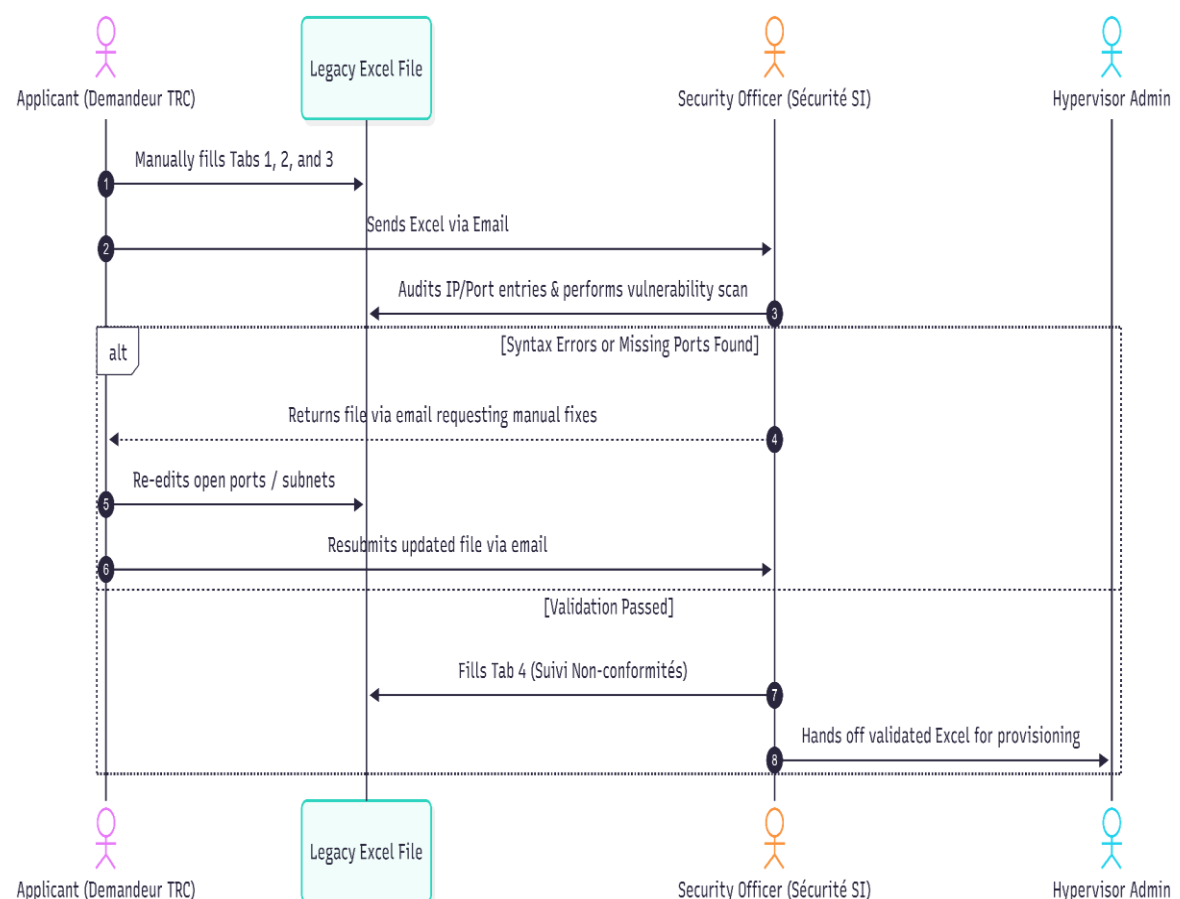


Figure 2: UML Sequence Diagram of the As-Is Manual VM Request Hand-off Process

2.2. Critical Diagnosis and Limitations

A diagnostic audit of historical publication forms revealed recurring operational failure points:

- **High Rejection Rates:** Over 40% of submitted requests were returned due to incomplete network port definitions or syntax errors in IP addresses.
- **Security Omissions:** Standard requesters frequently requested open HTTP/HTTPS access without specifying required DMZ network routing or TLS termination rules.
- **Lack of Field Protection:** Spreadsheets lacked cell-level access controls, allowing applicants to accidentally edit or overwrite security compliance fields reserved for *Sécurité SI*.

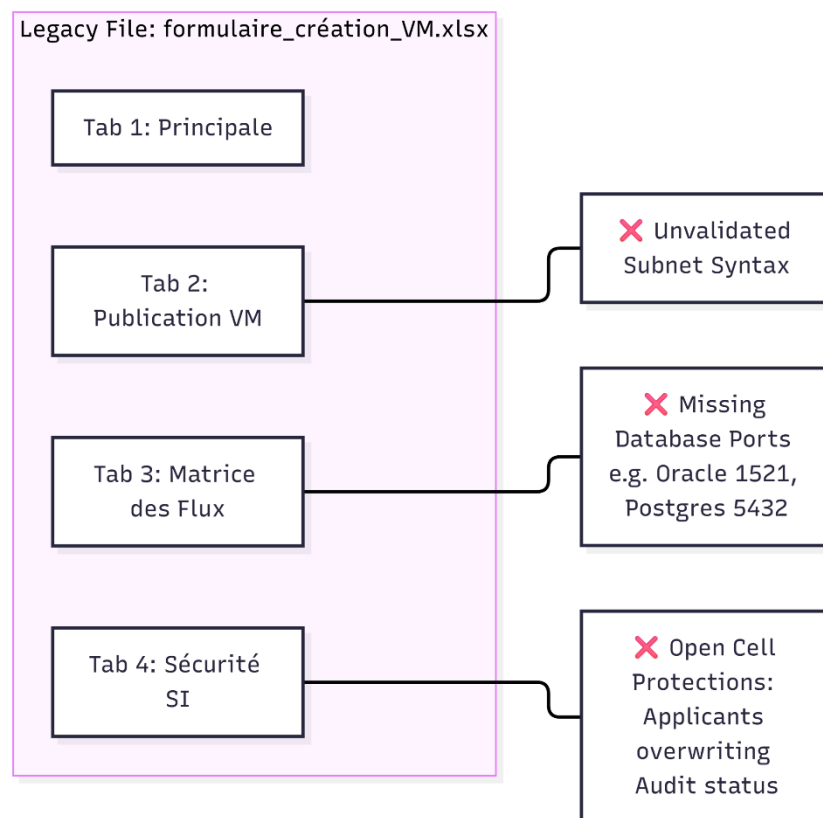


Figure 3: Misconfiguration Points Breakdown

2.3. Functional Requirements

The system functional requirements are organized into five core modules:

- **FR-01: Step-by-Step Form Wizard:** Provide a structured, multi-step interface that guides users through capturing metadata, technical VM configurations, software stacks, and network matrices.
- **FR-02: Real-Time Input Validation:** Perform client-side and server-side syntax checks on IPv4 addresses, subnet masks, port ranges, and DNS strings before form submission.

- **FR-03: Role-Based Security Audit Module:** Provide a dedicated dashboard for the *Sécurité SI* team to record non-conformities, evaluate DMZ placement, review vulnerability scans, and sign off on requests.
- **FR-04: Dynamic Multi-Tab Excel Generation Engine:** Export approved publication requests directly into the standardized multi-tab Excel template while preserving official layouts and formatting.

2.4. Non-Functional Requirements

- **NFR-01: Authentication & Access Control:** Role-Based Access Control (RBAC).
- **NFR-02: Data Integrity & Confidentiality:** Ensure strict read-only restrictions on security audit fields for standard requesters and encrypt network matrix data in transit and at rest.
- **NFR-03: Performance & Responsiveness:** Maintain rule inference response latency under 100 ms and complete multi-tab Excel file generation in under 2 seconds.
- **NFR-04: Auditability & Traceability:** Maintain immutable transaction logs for every request submission, modification, and security validation state transition.

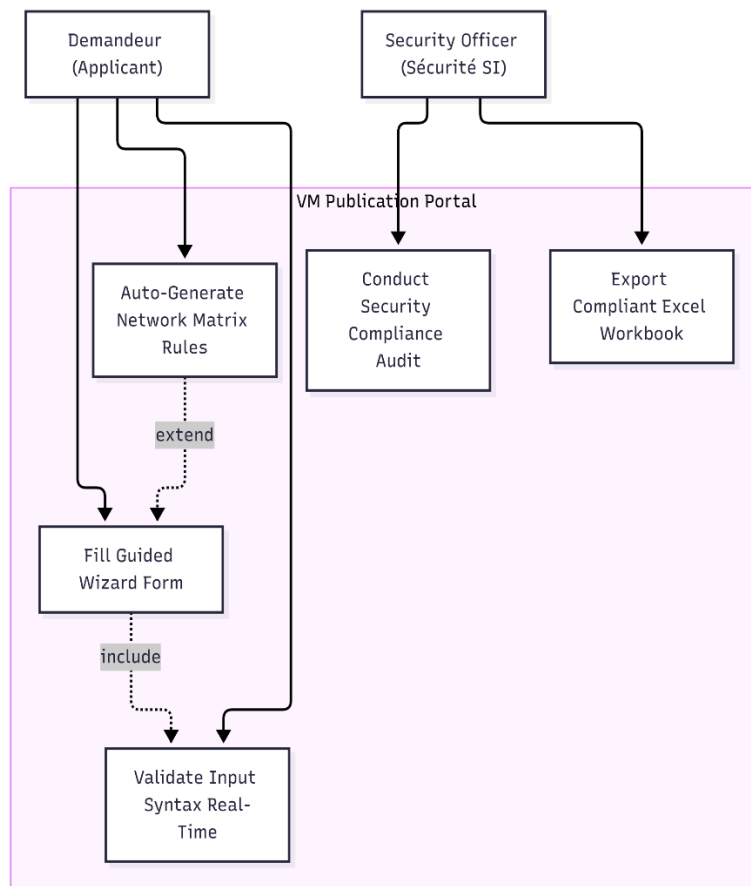


Figure 4: Global UML System Use Case Diagram

Chapter 3: System Design & Functional Architecture

3.1. Multi-Tab Form Pipeline Architecture

The platform streamlines network and security requests across four sequential functional tabs to ensure standardized data entry and prevent manual spreadsheet errors:

- **Tab 1 (Principale):** Captures project metadata, applicant contact information, host department, and administrative contacts.
- **Tab 2 (Publication VM):** Collects virtual machine specs, hostname conventions, operating system choices, and assigned IP addresses.
- **Tab 3 (Matrice Flux):** Specifies source-to-destination network communication rules (IPs, protocols, ports, and flow classifications).
- **Tab 4 (Sécurité SI):** Assesses cybersecurity compliance, backup frequencies, and system criticality.

3.2. Database Entity-Relationship (ER) Diagram

The relational database design normalizes form inputs into dedicated relational entities linked by a primary request identifier.

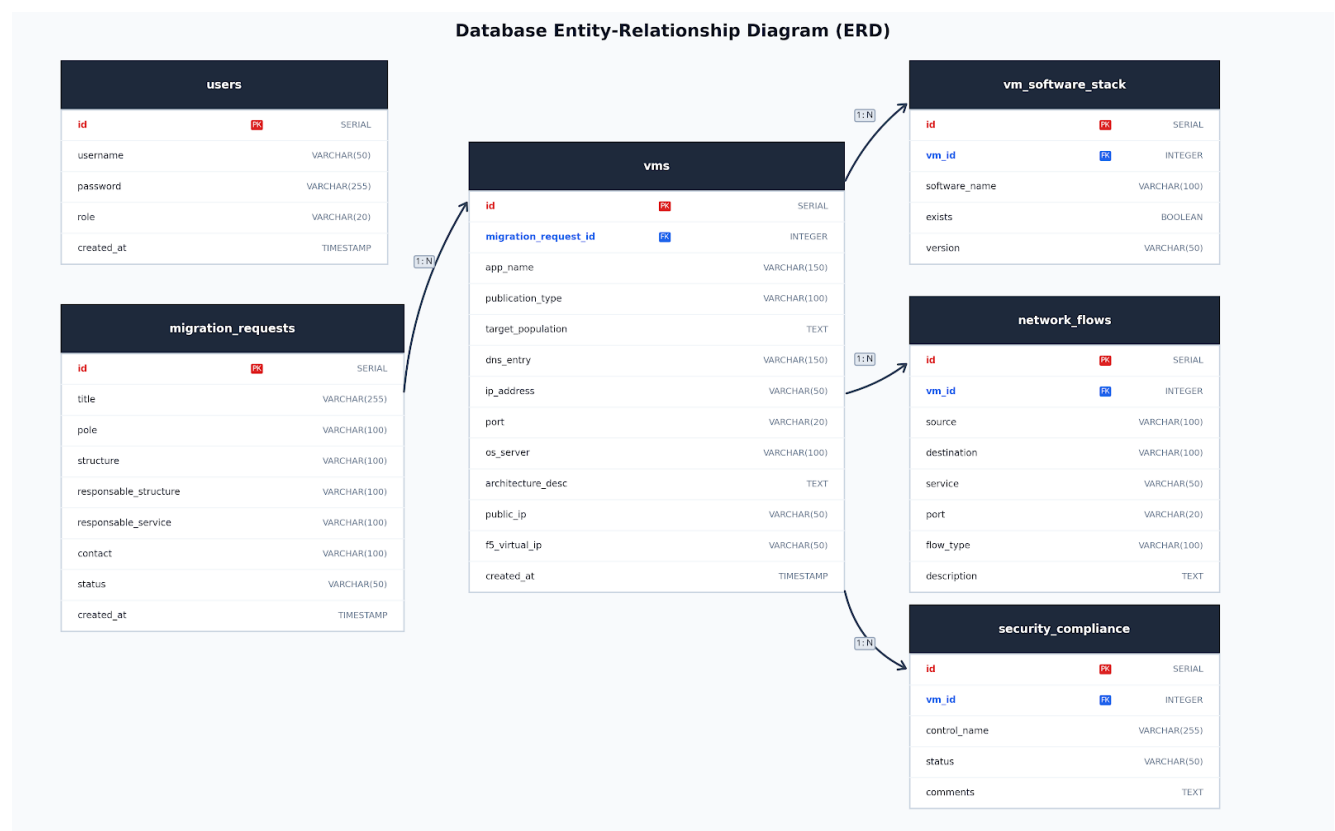


Figure 5: Database ER Diagram

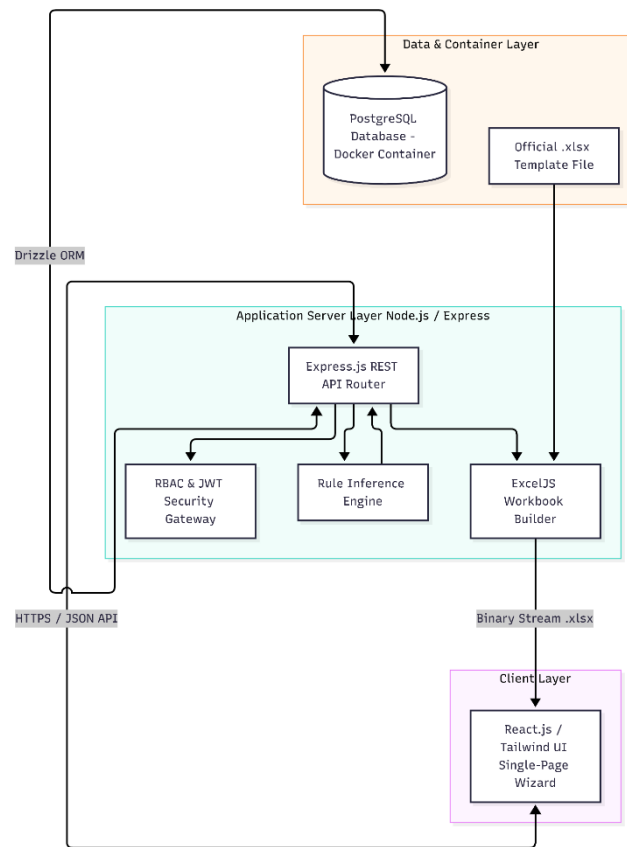


Figure 6: PERN System Architecture

3.3. Form Submission & State Lifecycle

Form processing moves through four defined states to ensure data validation prior to security validation and final export approval.

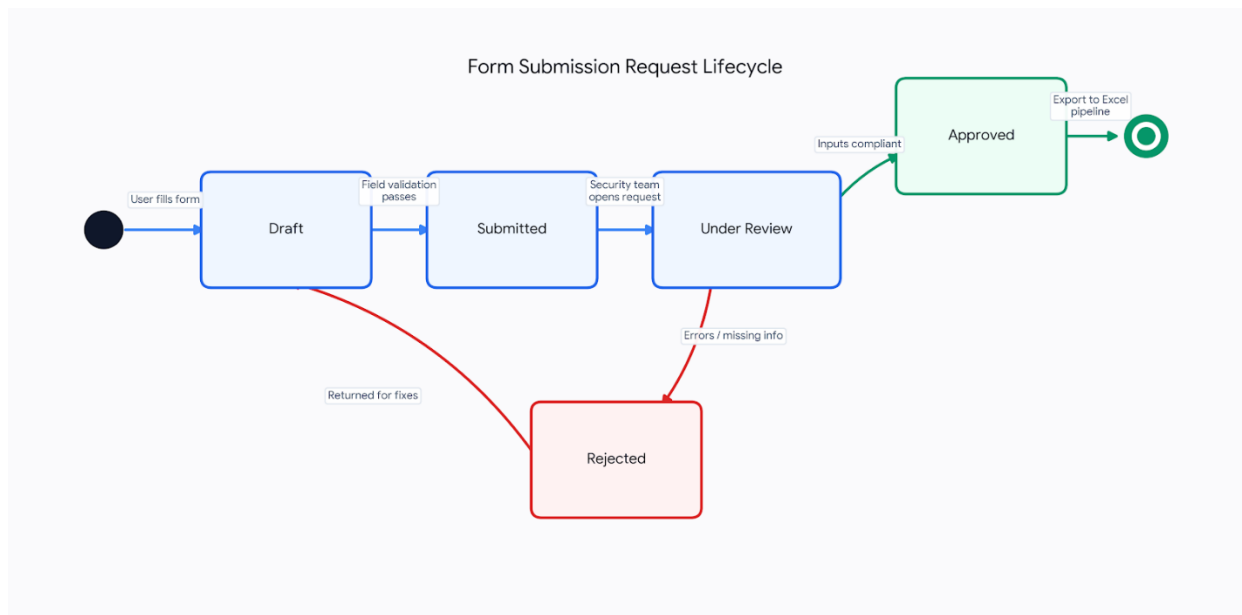


Figure 7: State Machine Diagram of form submission

Chapter 4: Implementation, Interface Design, and System Testing

4.1. Software Environment and System Configurations

The platform is built on the **PERN stack** (PostgreSQL, Express.js, React.js, Node.js) and engineered specifically for Sonatrach TRC (Direction Centrale IT). To ensure lightweight deployment and simple database management, PostgreSQL runs inside a dedicated **Docker** container, while the application server and client run on standard Node.js environments.

Architectural Layer	Technology Stack	Configuration Details
Frontend Framework	React.js / Tailwind CSS	Single-page wizard UI with dynamic modal navigation and client-side validation.
Backend REST API	Node.js / Express.js	Asynchronous JavaScript REST API handling routes, RBAC, and business logic.
Database System	PostgreSQL (<i>Dockerized</i>)	Isolated Docker container housing relational tables for requests, VMs, matrices, and audit logs.
Excel Export Engine	ExcelJS (Node.js)	Template-driven spreadsheet generator filling multi-tab .xlsx workbooks on the server.

Table 1: PERN Technology Stack Architecture & Configuration Details

4.2. Core Functional Interface Layouts (Step-by-Step UI & Security Portal)

The graphical user interface guides applicants and security reviewers through a structured sequence, starting from authentication to multi-tab data entry and export.

- **Authentication Portal & Main Dashboard**

- ✓ *Connexion Portail VM*: Enforces authentication via Sonatrach TRC IT credentials.
- ✓ *Demandes de Migration Dashboard*: Displays active migration requests, target structures/pôles, virtual machine badges, and row-level action triggers (**Voir / Gérer les VMs, + Saisir VM**).

- **Request Initialization Modal**

- ✓ Prompts users to define a title and operational description (e.g., *Migration Application Intranet TRC 2026*) to establish the parent request entity before attaching virtual machine technical sheets.

- **Tab 1: Principale (Demandeur TRC)**

- ✓ Collects organizational metadata including Pôle selection (e.g., *ALGER, Pôle Liquéfaction et Séparation*), target Structure, Structure Head, Service Owner, and primary contact details.

Figure 8: User Interface – Step 1: Applicant Metadata Form (Principale)

- **Tab 2: Publication VM (Fiche Technique & Stack)**

- ✓ Gathers technical hosting specifications: publication type (*Intranet/Extranet*), user base population, VM description, DNS entry, IP address, port, and Server OS (*Windows Server 2022, Red Hat Enterprise Linux 8.4*).
- ✓ Includes software stack configuration dropdowns (e.g., *Apache Struts, Apache Tomcat, Nginx, PostgreSQL*).

Figure 9: User Interface – Step 2: Technical Parameters and Software Stack (Publication VM)

- **Tab 3 : Informations liées au service (Flux & Architecture)**

- ✓ Captures high-level service architecture summaries and hosts the dynamic **Matrice des flux** table.
- ✓ Allows users to manually adjust or auto-populate network communication rules, specifying Source Subnet, Destination IP, Service, Port (e.g., 443, 22), Flow Type (*Flux applicatif Web*, *Flux d'administration*), and operational description.

Figure 10: User Interface – Step 3: Network Flow Matrix Generator (*Matrice des Flux*)

- **Tab 4: Suivi des non-conformités (Sécurité SI)**

- ✓ Captures public/internal network parameters (Public IP, Internal IP, F5 Virtual IP, Publication environment [DEV/PROD], last update timestamp).
- ✓ Provides an 8-point security compliance audit checklist (*DMZ Placement*, *Antivirus Installation*, *TLS Certificate*, *Authenticated Vulnerability Scan*, *Web Vulnerability Scan*, *Security Configuration Scan*, *Source Code Review*, and *Principle of Least Privilege [PoLP] compliance*).

Figure 11: User Interface – Step 4: Security Compliance Audit Module (*Sécurité SI*)

- **VM Management & Individual Export Action**

- ✓ Displays a consolidated modal view of all virtual machines associated with a specific migration request.
- ✓ Provides direct action buttons to edit parameters, delete records, or trigger immediate single-VM Excel generation via the **Export Excel** pipeline.

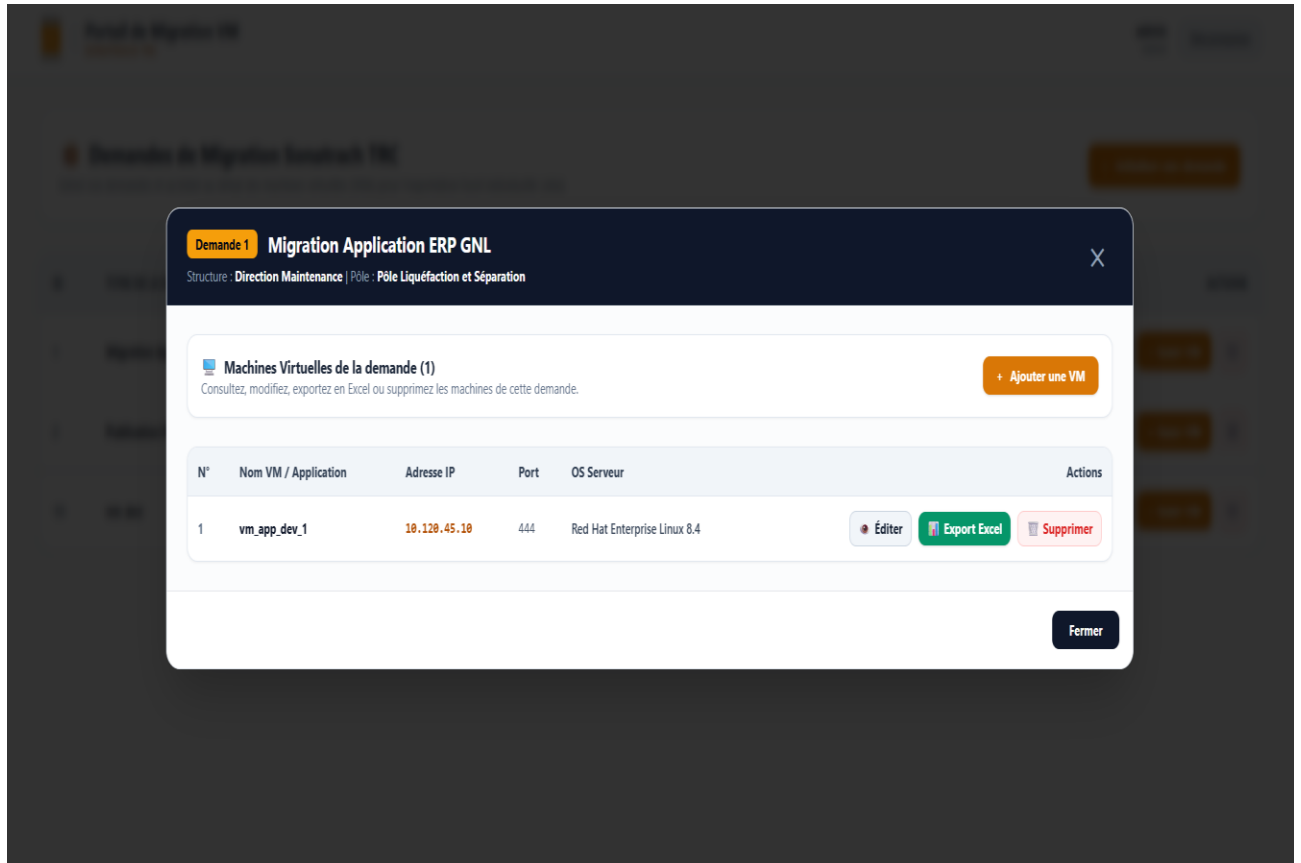


Figure 12: User Interface – Request Management Modal and Direct Excel Export Action

4.3. Excel Generation Engine Integration

Upon selecting **Export Excel**, the Express.js backend invokes **ExcelJS** to read the official enterprise template ([formulaire_création_VM.xlsx](#)), populate its worksheets with database records from PostgreSQL, and stream the generated workbook back to the client.

4.4. Evaluation, Tests, and Validation Results (Parsing & Rule Accuracy)

The system underwent testing across the PERN architecture, verifying database container communication, API response speeds, state transition enforcement, and **ExcelJS** spreadsheet output accuracy.

Test ID	System Module	Test Scenario	Expected Outcome	Result
TC-01	Authentication & RBAC	Submit valid Sonatrach IT credentials to Express endpoint.	Return signed JWT; render <i>Demandes de Migration</i> dashboard.	PASS
TC-02	Dockerized DB	Perform CRUD operations on PostgreSQL container.	Container handles query loads with zero lost connection drops.	PASS
TC-03	Input Validation	Leave required <i>Pôle</i> field unselected in Tab 1.	Block wizard step advancement; display UI error message.	PASS
TC-04	Rule Logic	Select <i>Red Hat Enterprise Linux 8.4 & PostgreSQL</i> .	Client auto-populates network flow entries for ports <i>22</i> and <i>5432</i> .	PASS
TC-05	Security Audit	Toggle audit check status to <i>Non valide</i> in Tab 4.	Record non-conformity in DB and append auditor comments.	PASS
TC-06	ExcelJS Pipeline	Trigger full multi-tab workbook generation via ExcelJS backend route .	Stream perfectly formatted 4-tab <i>.xlsx</i> matching enterprise layout.	PASS

Table 2: System Validation & Test Suite Execution Results

Conclusion

This internship report presents the analysis, design, implementation, and testing of an automated Virtual Machine (VM) Migration and Publication Portal for Sonatrach TRC (Direction Centrale IT). By replacing manual spreadsheet exchanges with a modern **PERN stack** (PostgreSQL inside a Docker container, Express.js, React.js, Node.js) web architecture, the platform streamlines the technical request lifecycle across operational divisions.

System Achievements & Operational Impact

- **Unified Single-Page Wizard:** Replaced manual data entry with a guided 4-step wizard (*Principale*, *Publication VM*, *Matrice Flux*, and *Sécurité SI*), guaranteeing field-level data validation before submission.
- **Containerized Relational Infrastructure:** Deployed PostgreSQL within a **Docker** container, establishing strict entity relationships between parent migration requests, virtual machine records, network flow matrices, and audit compliance logs.
- **Dynamic Network Matrix Engine:** Eliminated manual network flow configuration errors by dynamically generating 6-column rule mappings (Source, Destination, Service, Port, Flow Type, Description) based on selected infrastructure components.
- **Automated Excel Export Pipeline:** Integrated **ExcelJS** on the Express.js backend to read Sonatrach's official `formulaire_création_VM.xlsx` template and stream formatted multi-tab `.xlsx` workbooks directly to users.

Perspectives & Future Enhancements

- **Firewall Automation (NetOps Integration):** Interfacing the network matrix engine via REST APIs directly with enterprise firewall management systems (e.g., Palo Alto Networks, Fortinet) to push approved rules into active security policies automatically.
- **Active Directory / LDAP SSO Integration:** Upgrading local JWT authentication to Sonatrach's centralized Active Directory via LDAP/OAuth2 for single sign-on across departments.

References

- Codd, E. F. (1970). A relational model of data for large shared data banks. *Communications of the ACM*, 13(6), 377–387. <https://doi.org/10.1145/362384.362685>
- Drizzle Team. (2026). *Drizzle ORM documentation: TypeScript SQL ORM* [Software documentation]. <https://drizzle.team>
- ExcelJS Project. (2025). *ExcelJS: Excel workbook manager for Node.js and browser* [Computer software]. GitHub. <https://github.com/exceljs/exceljs>
- Express Team. (2026). *Express: Fast, unopinionated, minimalist web framework for Node.js* [Software documentation]. <https://expressjs.com>
- Fielding, R. T. (2000). *Architectural styles and the design of network-based software architectures* [Doctoral dissertation, University of California, Irvine]. ProQuest Dissertations & Theses Global.
- SONATRACH TRC. (2023). *Formulaire technique de publication et de mise à disposition des ressources* (Rev. 1.0) [Unpublished internal document]. Direction Générale / Direction Centrale Informatique et Système d’Information.